

Thesis Defense:
Invariant measures for the stochastic heat equation with
no drift term

Nicholas Eisenberg

March 29, 2022

The stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x), & t \geq 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The Stochastic Heat Equation is interpreted as follows:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(u(s, y))W(ds, dy)$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

- A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

The noise term $\dot{W}$

- $W = \{W(\phi) : \phi \in \mathcal{S}(\mathbb{R}^d)\}$ is a centered Gaussian process uniquely defined through the following covariance functional:

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$. We have the following nomenclature.

- *Space-time white noise*: $\Gamma(dx) = \delta_0(dx)$.
- *Time white and spatially colored*: $\Gamma(dx) = f(x)dx$ where f is a non-negative and non-negative definite function referred to as the *Correlation function* and its Fourier transform,

$$\hat{f}(\xi) := \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-ix \cdot \xi) f(x)dx,$$

is referred to as the *spectral density*.

Proposition (DaPrato and Zabczyk [7, Propsition 4.19])

A space-time white noise is the distributional derivative of a Brownian sheet, $B(t, x)$.

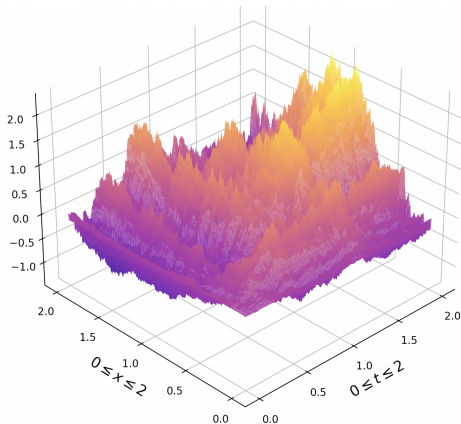


Figure: Brownian Sheet Simulation (Walsh [8]):

$$B(t, x) \sim B(t - \delta_t, x) + B(t, x - \delta_x) - B(t - \delta_t, x - \delta_x) + \text{normal}(0, \delta_t \delta_x)$$

$$B(t, 0) = B(0, x) = 0 \text{ for all } t, x > 0.$$

Existence and uniqueness with space-time white noise

- A necessary condition for global existence, *Dalang's condition* [6], is the following:

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) < \infty, \quad \forall t > 0.$$

- Direct calculation yields that,

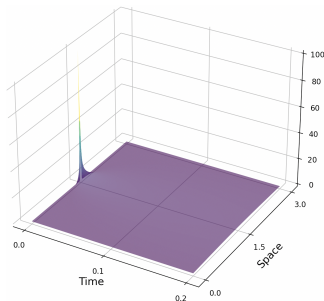
$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) = \frac{\Gamma(d/2)}{2(2\pi)^d} \int_0^t \frac{1}{s^{d/2}} ds,$$

which forces,

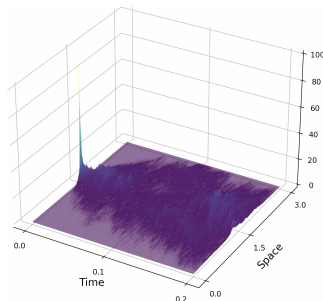
$$d/2 < 1 \quad \Longleftrightarrow \quad d < 2.$$

- Thus a global space-time white noise solution may only exist when the spatial dimension is 1.

Large Peaks: $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ with $u_0(x) = \delta_0(x)$
 driven by a space-time white noise



$\lambda = 0$

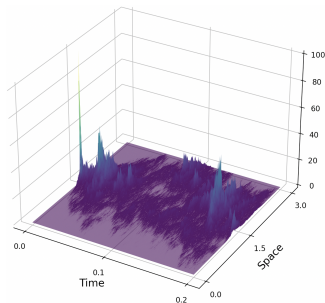


$\lambda = 1$

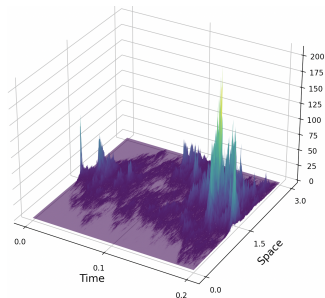
- Simulations done by using a finite difference scheme [5].
- We partition (space) $[0, 3]$ into n_x pieces of size $\delta_x = 3/n_x$.
- We partition (time) $[0, .2]$ into n_t pieces of size $\delta_t = .2/n_t$.

$$u(t + \delta_t, x) = u(t, x) + n_x^2 \delta_t (u(t, x + \delta_x) + u(t, x - \delta_x) - 2u(t, x)) \\ + n_x \lambda u(t, x) \text{normal}(0, \delta_t \delta_x).$$

Intermittency of $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ driven by space-time white noise



$\lambda = 1.8$



$\lambda = 2$

● Notice the enlargement of the peaks as λ increases. We have the following theorem.

Theorem (Chen and Dalang [1, Theorem 2.4])

The second moments of the stochastic heat equation driven by space-time white noise grow exponentially in time.

Will the second moments ever be bounded in time?

- Several slides below will require the second moments to be bounded in time:

$$\sup_{t \geq 0} \|u(t, x)\|_2 < \infty, \quad \text{where } \|X\|_p = [\mathbb{E}(|X|^p)]^{1/p}.$$

- This is possible, however this will require us to only consider $d \geq 3$, and therefore we must only consider a spatially colored noise.
- Recall from above, that the covariance of a spatially colored noise is given by the following:

$$\begin{aligned} \mathbb{E}[W(\phi)W(\psi)] &= \int_0^\infty ds \int_{\mathbb{R}^d} dx f(x)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) f(x - y) \psi(s, y) \\ &= (2\pi)^{-d} \int_0^\infty ds \int_{\mathbb{R}^d} d\xi \hat{f}(\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)} \end{aligned}$$

Existence and Uniqueness with a Colored Noise

Theorem (Chen and Huang [2, Theorem 1.2])

There exists a unique global mild solution under the following:

- $b : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a $L_b, L_0 > 0$ so for all $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0.$$

- The initial measure, $\mu(dx)$ (resp. $\varphi(x)dx$), satisfies for some $a > 0$,

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty \quad \left(\text{resp.} \quad \int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty \right),$$

where $|\mu| = \mu^+ + \mu^-$ is the *Hahn decomposition*.

- The spectral density, $\hat{f}$, satisfies *Dalang's condition* for some $\beta > 0$:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

Admissible Weight

Since we work in the whole space $\mathbb{R}^d$, we consider weights, ρ such that

- ▶ $\rho \in L^1(\mathbb{R}^d)$ is positive, continuous and bounded,
- ▶ for any $T > 0$, there exists a $C(T) > 0$ such that

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho)(x)}{\rho(x)} \leq C(T).$$

We then work in the weighted Hilbert Space, $L^2_\rho(\mathbb{R}^d)$, with norm defined as

$$\|\phi(\cdot)\|_{L^2_\rho(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} \phi(x)^2 \rho(x) dx.$$

Some examples of admissible weights include the following:

$$\frac{1}{1 + |x|^a}, \quad a > d \quad \text{and} \quad \exp(-a|x|), \quad a > 0.$$

Intermittency

We define the upper and lower Lyapunov exponents

$$\underline{m}_2 := \liminf_{t>0} \frac{1}{t} \log \left(\mathbb{E}(|u(t, x)|^2) \right), \text{ and } \overline{m}_2 := \limsup_{t>0} \frac{1}{t} \log \left(\mathbb{E}(|u(t, x)|^2) \right).$$

The solution is said to be *fully intermittent* when

$$\inf_{x \in \mathbb{R}^d} \underline{m}_2 > 0.$$

We say a *phase transition* happens when there exists $\underline{\lambda} \leq \overline{\lambda}$ such that

$$\begin{cases} \sup_{x \in \mathbb{R}^d} \overline{m}_2 = 0 & L_b < \underline{\lambda} \\ \inf_{x \in \mathbb{R}^d} \underline{m}_2 = 0 & \overline{\lambda} < l_b \end{cases}$$

Bounded Moments

Under the assumptions for existence and uniqueness, we have the following moment bound

$$\|u(t, x)\|_p \leq C[1 + J_+(t, x)]H(t; \gamma_p)^{1/2}, \quad \gamma_p = 32pL_b^2,$$

where $t \mapsto H(t, \gamma_p)$ is nondecreasing. Following a strategy laid out by Daprato and Zabczyk, we wish to prove that

$$\sup_{t \geq 0} \mathbb{E} \left(\|u(t, x)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Hence, a phase transition must occur. By Theorem 1.3, this forces us to require $d \geq 3$ and $\Upsilon(0) < \infty$. If we further assume that L_b is small then we may also say

$$\|u(t, x)\|_p \leq C[1 + J_+(t, x)].$$

Invariant Measures

Consider a measure space $(E, \mathcal{E})$ and denote $\mathcal{M}_1(E)$ to be the set of probability measure on this space.

Definition (Invariant Measure)

A measure $\eta(\cdot) \in \mathcal{M}_1(E)$ is invariant for transition functions $\{P_t(x, \Gamma)\}_{t \geq 0}$ if for any $\Gamma \in \mathcal{E}$ we have that

$$\eta(\Gamma) = \int_E P_t(x, \Gamma) \eta(dx) \quad \text{for any } t \geq 0$$

Recall from Calculus that the average of a function $f : [a, b] \rightarrow \mathbb{R}$ is defined as

$$\text{AVG}(f) = \frac{1}{b-a} \int_a^b f(x) dx$$

Thus, $\eta(\Gamma)$ is the average the family of probability measures $\{P_t(x, \Gamma)\}_{x \in E}$ for any $t \geq 0$.

Theorem: Existence of an Invariant measure

A processes $u(t, \cdot)(\omega)$ is bounded in probability in $L^2_\rho(\mathbb{R}^d)$ if

$$\forall \epsilon > 0, \exists R > 0 : \forall t > 0, \mathbb{P} \left\{ \|u(t, \cdot)\|_{L^2_\rho(\mathbb{R}^d)} \geq R \right\} \leq \epsilon.$$

Tessitore and Zabczyk proved the following:

Theorem

Suppose $d \geq 3$ and that there exists a $p \in [1, d/(d+2))$ such that the spatial correlation density $\gamma \in L^p(\mathbb{R}^d)$. Suppose that there exists two weights, ρ and $\hat{\rho}$ and an element $\phi \in L^2_\rho(\mathbb{R}^d) \cap L^2_{\hat{\rho}}(\mathbb{R}^d)$ such that

$$\frac{\rho}{\hat{\rho}} \in L^1(\mathbb{R}^d) \quad \text{and} \quad u^\phi(t, \cdot) \text{ is bounded in probability in } L^2_{\hat{\rho}}.$$

Then there exists an invariant measure in $L^2_\rho(\mathbb{R}^d)$.

Sketch of the Proof of Existence

It has been shown by DaPrato and Zabczyk that $P_t(\phi, \Gamma) := \mathcal{L}(u^\phi(t, \cdot))(\Gamma)$ defines a Markov transition function on $(L_\rho^2(\mathbb{R}^d), \mathcal{L})$ where,

$$\mathcal{L}(u^\phi(t, \cdot))(\Gamma) = \mathbb{P} \left[\omega \in \Omega : u^\phi(t, \cdot)(\omega) \in \Gamma \right], \quad \Gamma \in \mathcal{L}.$$

We then prove for any $\epsilon \in (0, 1)$ and $R > 0$, the existence of a compact set, $\mathcal{K} \in \mathcal{L}$ such that

$$\mathcal{L}(u^\phi(t, \cdot))(\mathcal{K}) \geq (1 - \epsilon) \mathbb{P} \left[\omega \in \Omega : \left\| u^\phi(t - 1, \cdot)(\omega) \right\|_{L_\rho^2(\mathbb{R}^d)} < R \right], \quad t \geq 1.$$

If we assume $u^\phi(t, \cdot)$ is bounded in probability in $L_\rho^2(\mathbb{R}^d)$ then one can argue the existence of a sequence $\{T_n\}$ such that $0 < T_n \uparrow \infty$

$$\frac{1}{T_n} \int_{t_0}^{T_n+t_0} \mathcal{L}(u^\phi(t, \cdot))(\Gamma) \xrightarrow[\text{weakly}]{} \eta(\Gamma) \quad \forall \Gamma \in \mathcal{L},$$

and moreover, η is invariant in $L_\rho^2(\mathbb{R}^d)$.

Tessitore and Zabczyk's Further Assumption

If $B := \sup_{t \geq 0} \mathbb{E} \left(\|u(t, x)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty$ then by Chebyshev's inequality

$$\mathbb{P}[|u(t, \cdot)|_\rho \geq R] \leq R^{-2} \mathbb{E}|u(t, \cdot)|_\rho^2 \leq R^{-2} B < \epsilon, \quad \text{for any } R \geq \sqrt{B/\epsilon}.$$

Theorem

Suppose that $d \geq 3$ and that $L < L_b^{-2}$ where

$$\tilde{\Gamma} := \left| \widehat{\gamma^{1/2}} \right| * \left| \widehat{\gamma^{1/2}} \right| \quad \text{and} \quad L := \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \int_{\mathbb{R}^d} \tilde{\Gamma}(\zeta) |\zeta|^{2-d} d\zeta < \infty.$$

Then for any admissible weight ρ , we have that

$$\sup_{t \geq 0} \mathbb{E} \left(\left\| u^1(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Moment Bound: Measure Valued Initial Conditions

Theorem

Suppose that the initial measure, μ , and spectral measure, $\hat{f}$:

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < \infty \quad \forall t > 0 \quad \text{and} \quad \Upsilon(0) < \infty. \quad (0.1)$$

Then if $\mu(dx) = \phi(x)dx$ with $\phi \in L^2_\rho(\mathbb{R}^d)$,

$$\sup_{t \geq 0} \mathbb{E} \left(\left\| u^\phi(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Further, if μ has no density but we strengthen (0.1) to include

$$\int_0^s dr \, r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2) < \infty$$

then for any $t_0 > 0$,

$$\sup_{t \geq t_0} \mathbb{E} \left(\left\| u^\mu(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Invariant Measures: Measure Valued Initial Conditions

We wish prove the existence of an invariant measure for measure valued initial conditions such as the Dirac delta measure, $\delta_0(\cdot)$. Our plan is to consider

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

and then restart our system. In other words, we will consider

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2}\Delta_x v(t, x) = b(x, v(t, x))\dot{W}(t, x) \\ v(0, \cdot) = u(t_0, \cdot) \end{cases}.$$

With the absence of noise, we know that $v(t, x) = u(t + t_0, x)$ and we wish to show that this property holds with the addition of noise. We also know that $\sup_{t \geq 0} \|u(t + t_0, \cdot)\|_{L^2_\rho(\mathbb{R}^d)} < \infty$. Thus we need to figure out how extend the known proof to handle random initial conditions.

References



L. Chen and R. Dalang (2015)

Moments and growth indices for the nonlinear stochastic heat equation with rough initial conditions

The Annals of Probability 43, 6, 3006–3051



L. Chen and J. Huang

Comparison principle for stochastic heat equation on $\mathbb{R}^d$

Ann. Probab., 47.2 (2019) 989–1035.



L. Chen, Y. Hu, D. Nualart (2019)

Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$.

Stochastic Processes and their Applications, 129 5073–5112.



L. Chen and K. Kim (2019)

Nonlinear stochastic heat equation driven by a spatially colored noise: moments and intermittency.

Acta Mathematica Scientia 39B(3), 645–668



A. Davie and J. Gaines (2000)

Convergence of numerical schemes for the solution of parabolic stochastic